

More than just managerial self-efficacy: conceptualizing and predicting top managers' means efficacy about the organization under extreme events

Predicting
MME of the
organization

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Abstract

Purpose – This study extends the concept of managerial efficacy to include managerial means efficacy (MME) attributed to the utility and quality of means external to managers for performing a task. Focusing on its antecedents, the authors theorize and empirically test MME sourced from the organization (MMEO) and situate the examination under extreme events.

Design/methodology/approach – This study uses a 2016 national survey of 892 top managers in 273 US largest transit agencies covering 82% of the entire population. Participants report their confidence for their organization to perform effectively under various extreme events. The survey data are matched with data from various institutional data sources to capture organizational characteristics, operations and experience with extreme events.

Findings – Findings suggest that organizational-level factors influence MMEO. Specifically, organizational slack and relationship management of key jurisdictional stakeholders positively predict MMEO, whereas political fragmentation is negatively associated with MMEO.

Practical implications – Organizations can bolster MMEO, hence, managerial efficacy through developing organizational slack and engaging in relationships building with jurisdictional stakeholders. Those initiatives have particular importance for those with boundary-spanning service areas.

Originality/value – The study advances understanding of managerial efficacy by directing attention to means external to managers' self. It also brings clarity to the notion of "confident managers" or "managerial confidence" broadly applied in previous studies. Findings provide insights about capacity-building interventions to build managerial efficacy through improving external means, circumventing the need to alter self-efficacy that is typically stable and resistant to change.

Keywords Managerial efficacy, External sources of efficacy beliefs, Managerial means efficacy, Means efficacy of organization, Extreme events

Paper type Research paper

Introduction

Managers and their organizations face ever-accelerating rates of changes in their environments. A constantly turbulent environment not only challenges managers' competence, knowledge and skills, but also tests overall efficacy beliefs that provide psychological capacity to cope with demands (Hannah *et al.*, 2008). When it comes to the specifics of managerial efficacy, studies



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have predominantly considered managerial self-efficacy, citing antecedents such as enactive mastery, intelligence, learning capability and goal orientation (Appelbaum and Hare, 1996; Bandura, 2000; Brockner and James, 2008).

Despite their unambiguous focus on individuals' personal abilities, scholars commonly measure managerial self-efficacy by asking managers about their subjective assessment of performing a specific task in a given situation (Eden *et al.*, 2010). This operationalization likely contradicts the inward-looking concept of self-efficacy, because it can capture not only the managers' beliefs about their own abilities, but also aspects of the situation that can facilitate or hinder performance, notably the external means available for carrying out the task (Eden *et al.*, 2010). Applying Eden's theory of means efficacy (2001), this study distinguishes two sources of managerial overall efficacy: (1) *self-efficacy* relating to a manager's appraisal of resources *internal* to the manager, such as the manager's talents, skills, expertise and endurance; and (2) *means efficacy* defined as the manager's appraisal of the quality and utility of means *external* to the manager. We focus on the second, which we call managerial means efficiency (MME).

Inclusion of external sources of managerial efficacy expands the efficacy concept and advances understanding of motivating efforts (Eden, 2001). Practically, the malleability of means efficacy enlarges our motivational toolboxes: without altering managerial self-efficacy, which is typically resistant to change (Luthans and Youssef, 2007), we can more effectively raise managers' means efficacy and enhance performance through capacity-building interventions. Example strategies include upgrades or reinforcement to improve problem-solving capability of a given means (e.g. a technology or management system), persuasive communication to foster beliefs about its efficacy, as well as training on effective use of the means (Chen *et al.*, 2009). To achieve the latter, it is necessary to identify the antecedents of MME. Eden (2001) notes various sources of MME, ranging from organization, tools, technology to teams. For a conceptual focus, we settle on MME sourced from the organization (MMEO) as a vehicle to examine the antecedents, because managers' ability to perform, get things done and attain outcomes are largely done through their organization (Eden, 2001; Pfeffer, 1992).

Lacking *a priori* framework on external sources of managerial efficacy, we find the entry from Ajzen's theory of planned behavior to conceptualize MMEO. This study dissects two dimensions of MMEO: (1) internal-facing perceived organizational capacity; (2) external-facing perceived controllability. Since tasks vary in the levels of means-dependence, we apply our new approach to one setting with high means-dependence: extreme events. Previous studies broadly note that managers have to marshal a vast array of personnel and resources and coordinate with myriad actors to produce desired outcomes under extreme events (Kapucu and Hu, 2016; McEntire, 2002; Pramanik *et al.*, 2015; Ulmer, 2001). Contingencies created by events and actors outside their organization heightens managerial reliance on external means to reduce uncertainty and ensure task performance (Hickson *et al.*, 1971; Pfeffer, 1992).

We focus on MMEO in transit agencies under the influence of extreme events for several reasons. First, transit organizations vary substantially in terms of structure and operations, level of contracting and political control. The wide spectrum of these organizations provides an important source of variation and potential for generalizability. Second, transit systems are complex, visibly exposed to environmental shocks and vary in terms of political authority and independence. In this research, extreme events impact all aspect of transit systems and operations making managerial efficacy a critical psychological resource to attenuate the event impacts (Hannah *et al.*, 2009). Finally, transit agencies are responsible for services across jurisdictional boundaries and operate in a multilevel government system, all of which reinforce their dependence on external means to perform complex and interrelated tasks during extreme events. Our approach and focus on transit agencies provide the appropriate means-dependent context to develop and test hypotheses about explanatory factors of MMEO.

We now turn to conceptualizing MMEO and describing the means-dependent context of extreme events. We test hypotheses using a 2016 national survey of top managers in the largest fixed-route US transit agencies merged with institutional data from various sources. The results generate new insights for understanding MMEO and spark further questions of research.

Theoretical framework

Conceptualizing MMEO

This study integrates the theory of means efficacy and theory of planned behavior to develop a theoretical framework for MMEO. Eden's theory of means efficacy accounts for managers' operating contexts by adding external sources of efficacy beliefs. Means efficacy is defined as one's belief in the utility of the means available for performing a job. It has similar motivating effects as self-efficacy because it elevates performance expectations and provides strong impetus for individuals to productively cultivate and utilize means to enhance performance (Eden *et al.*, 2010). Just as means exists independently of the manager's self, managerial belief in external means is distinct from the belief in self. The motivational power of high self-efficacy ("I am an experienced and competent manager") can be neutralized by low means efficacy ("The technology is ill-designed for the task").

Eden (2001) notes several sources of means efficacy, including organization, technology and tools, peers and work teams. Organizations are the first and foremost rational tools (Eden, 2001) to coordinate efforts from a collection of actors to reach a common goal. We propose that MMEO captures the extent to which MME is informed and shaped by managers' understanding of the organization's potential. The sense of having a well-equipped organization inspires MMEO and sparks efforts, whereas perceiving an underprepared organization can diminish MMEO and discourage efforts.

The theory of planned behavior provides an appropriate entrée to dissect and explain MMEO. A key construct in the theory is perceived behavioral control (PBC) defined as individuals' perceived level of ease or difficulty of performing a given behavior (Ajzen, 1991). In situations that afford an individual complete control over the outcome, intention alone is sufficient to predict behavior. Conversely, in situations where individuals lack complete control, PBC acts as a critical predictor for action and achievement (Ajzen, 1991, 2002). PBC's emphasis of reliance on resources, personnel and tools external to an individual matches well with the central idea of means efficacy.

MMEO under extreme events

Eden (2001) holds that for tasks with an extraordinary demand for external means, means efficacy can overshadow self-efficacy in determining goal attainment. We anchor the examination of MMEO in a high means-dependent context: extreme events. Extreme events, such as weather disasters, disease outbreaks and technological breakdowns, push managers and their organization to the limit and severely test their abilities to expeditiously undertake a complex set of interrelated tasks. When extreme events hit, stakes are high and intentions to perform are strong. But control is limited, as risks and uncertainties abound and the organization must depend on external actors to produce outcomes. Extreme events have a way of exposing deficits in the organization's system that managers are otherwise unaware of (Christianson *et al.*, 2009; Zhang *et al.*, 2020). As a result, extreme events exemplify a high means-dependent task environment that exposes the role of MMEO.

Ajzen (2002) delineates two underlying dimensions of PBC: (1) perceived self-efficacy expressing individual beliefs of a person's skills or capabilities to perform a behavior; and (2) perceived controllability defined as individual beliefs about the extent to which performing

the behavior is up to the actor. These two dimensions reflect the internal and external causes of PBC and have been found to differ empirically from each other (Ajzen, 2002). Shifting the referent of efficacy beliefs from the individual to the organization, we conceptualize MMEO as varying along two dimensions: (1) *perceived organizational capacity* reflects top managers' beliefs about their organization's internal resources and capabilities to perform a task; and (2) *perceived controllability* indicates managerial beliefs about the level of control the organization has over task execution and outcome attainment.

Figure 1 summarizes the theoretical model. It identifies perceived organizational capacity and perceived controllability as two underlying dimensions of MMEO. Perceived controllability is further theorized to depend on input organizations and the broader political contexts. The empirical model will focus on how organizational slack, outsourcing, political fragmentation and management of relationships with jurisdictional actors predict MMEO.

Hypotheses

Perceived organizational capacity

Perceived organizational capacity captures manager appraisal of the enabling and constraining conditions internal to the organization for performance during extreme events. Despite the heterogeneity of threats, general capacities can be applied to a broad range of extreme events (Cimellaro et al., 2010), including robustness to withstand shocks, resourcefulness to apply information and resources, redundancy to provide for temporary loss, and rapidity to mobilize and deliver response. It is typical for managers to approach an extreme event asking whether their organization has the requisite capacity to satisfy potential demands. Sensing a discrepancy between the existing capability and the desired state would cast doubt in the managers' mind.

Organizational slack. Organizational slack refers to the resources available to an organization in excess of what is required for routine operations (Cyert and March, 1963). During an extreme event, slack permits an organization to accommodate changes and absorb shocks gracefully without rupturing in face of tremendous stress and demand (Bourgeois, 1981, p. 30). Slack resources afford organizations and managers the discretion and flexibility to make timely adjustments or initiate strategic changes in response to internal and external

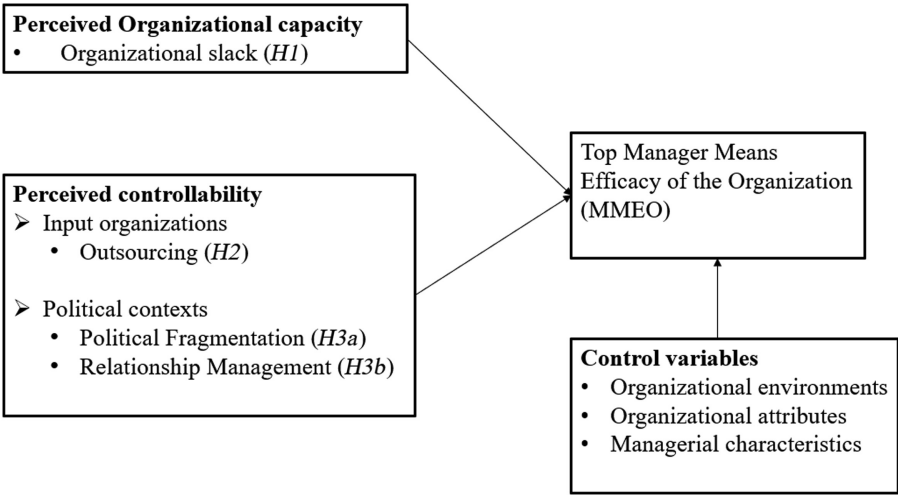


Figure 1.
Antecedents of MMEO
under extreme events

pressures. Resource-rich organizations are therefore better able to handle unexpected or surging demand and compensate for malfunctioning elements to enable continued operation (Stark, 2014). For instance, a transit agency might not run its vehicles at full capacity to allow for contingency dispatches, or it may store portable generators to sustain power in case of power failure.

The level of slack may vary by industry or sector, depending on product cycle, risk exposure and industry structure (Marlin, 2014; Wefald *et al.*, 2010). For example, slack may be lower in industries that have shorter product cycles and therefore lower margins (e.g. consumer electronics, food products and home appliances) than in those that are more technology intensive and have longer product cycles (e.g. energy or electronic equipment) (Bradley *et al.*, 2011). Additionally, slack may be higher in the service sector where there is a greater need to respond to rapid fluctuations in service demand. Because transit has characteristics of both longer-production cycles requiring high fixed capital investment levels and shorter production cycles due to variable ridership demands, it can be classified as a mixed service industry (Sharfman *et al.*, 1988) with relatively moderate slack needs. Recognizing the potential for industry differences, it is generally the case that pre-existing deficiencies or emerging errors become more tolerable and remediable at the organizational level when auxiliary resources are higher. Besides creating a critical buffer during an event, slack provides managers with resources to investigate problems and vulnerabilities and make anticipatory adaptations before events arrive (Miao *et al.*, 2018a).

H1. Organizational slack is positively associated with MMEO under extreme events.

Perceived controllability

Perceived controllability reflects managerial beliefs about the organization's control over the operation of external factors. Extreme events cut across political, geographical and functional boundaries wherein the involvement of external actors necessarily constrains decision making and implementation of the focal organization. Resource dependence theory identifies two primary sets of factors in an organization's environment: the set of organizations it interacts with and the larger political context (Evan, 1965). Input organizations and political contexts are of particular relevance. Input organizations provide various essential resources to the focal organization. Political contexts capture the complexity of the operating environment and required activities to implement a given course of action.

Input organizations. During extreme events, an organization needs to orchestrate coordinated response with all parties involved. This requires input organizations to communicate critical information, provide essential supplies and act in concert with the focal organization. Whether the focal organization can secure satisfactory cooperation from its input organizations will influence managers' perceived controllability.

Das and Teng (1998) distill two determinants of managerial beliefs in obtaining satisfactory cooperation: trust and control. *Trust* is the degree to which the focal organization holds a positive attitude toward the trustee's goodwill and competence. Two facets of trust emerge from this definition: (1) goodwill trust exists when the manager is assured that the input organization will care about its welfare and operate in good faith; (2) competence trust exists when the manager believes that the input organization has the required resources and competence to perform certain tasks. *Control* expresses the extent to which the manager believes regulatory and control arrangements can adequately ensure predictability and accountability of the input organization. Perceived control is possible when managers have faith in the input organization's goodwill and competence and/or when sufficient mechanisms are in place to preclude self-interest and opportunistic behavior (Ajzen, 2002).

Outsourcing: Outsourcing occurs when organizations delegate functions of goods and service provision to a third party. Reliance on third-party organizations can undermine

managerial perception of control in the dyadic relationship with contractors for several reasons. Extreme events amplify the costs of action, and since contractors are fundamentally guided by economic efficiency, they may be less likely to carry out the contracted activities due to escalating costs. Breaches of contracts and price gouging often occur during extreme events (Egan, 2010; Gotham, 2012). Equally questionable is the competence of contractors to perform during extreme events. The preparedness of contractors is conspicuously lacking in the current crisis management paradigm, as the protocols for preparation and response to extreme events among private firms is a recent development and has yet to be widely diffused (Tierney, 2014).

The effects are exacerbated as the organization has limited control over its contractors during extreme events. Agency theory suggests that the principal can control agent behavior through behavioral or outcome control (Eisenhardt, 1989). However, the non-routineness, unpredictability and complexity of extreme events undercut the feasibility of predetermined processes and procedures. Organizational decision making resembles an organized anarchy, with problematic preferences, unclear technology and fluid participation (Comfort, 1993). Both situations lead to ambiguity and uncertainties of means and goals, severely hampering the feasibility of either behavioral or outcome control. Further complicating the matter are the legal provisions that excuse non-performance on the grounds of impossibility, infeasibility and frustration of purposes (Egan, 2010).

In summary, there are strong reasons to believe that top managers lack sufficient trust and control of third-party contractors during extreme events. Their perceived controllability and hence MMEO would be negatively associated with the level of outsourcing.

- H2. The level of organizational outsourcing is negatively associated with MMEO under extreme events.

Political contexts. The study covers two aspects of political contexts: political fragmentation and relationship management with stakeholders.

Political fragmentation: Political fragmentation reflects the extent to which the political authority, expertise and resources required for crisis response are distributed across jurisdictions. As political fragmentation increases, the scope of a problem widens and the demand for coordination and integration intensifies, leading to lower levels of perceived controllability. Organizations acting in dispersed locations receive partial information from various sources at different levels of governments about the unfolding event. In high means-dependent contexts, the difficulties of information flow and collective sense-making quickly escalate when greater numbers of jurisdictions are involved (Comfort, 2002). Jurisdictional fragmentation makes it difficult to establish and maintain clear lines of control during extreme events because authority must be negotiated between multiple, potentially conflicting, actors in a governance network, each with their own power advantages and dependencies (Ansell and Gash, 2008; Fleming *et al.*, 2015; Pfeffer, 1992) and because of the ever shifting nature of these events across jurisdictions (Abbasi and Kapucu, 2016).

Coordination is further plagued by the diversity of norms, routines and standards of operation among the various jurisdictions. Divergent implementation strategies and tactics for problem-solving produce conflicting resource allocation protocols and processes that cause disputes during crisis response (Rosenthal and Kouzmin, 1997). Because those differences and capacity disparities add an extra layer of operational complexity, political fragmentation can further increase a manager's uncertainty about the organization's ability to control the course of action.

- H3a. The degree of political fragmentation is negatively associated with MMEO under extreme events.

Relationship management: Organizational effectiveness to manage relationships in the multijurisdictional context also matters to MMEO. Effectiveness is achieved when the

organization is capable of creating, maintaining and advancing positive relationships with key stakeholders to coordinate task performance.

Investment in building and managing relationships with other jurisdictional actors necessitates repeated and constant interactions, allowing organizations to learn about each other's processes, competencies and constraints, and develop common perspectives on activities required for joint response. The interacting parties also benefit from a better appreciation of their interdependence, raising the chance for anticipatory or synchronous mutual adaptation to emerging contingencies (Ulmer, 2001). Additionally, engagement with various jurisdictions helps the focal organizations build critical knowledge about who knows what, who is in charge of which functions and who can help with which problems. The interactions embed the focal organization in a network that builds trust and invests in social capital resources which can be converted and applied as assistance during a crisis response (Kapucu and Hu, 2016; Lin, 1999, 2001). A solid working relationship can compensate for a lack of formal coordination channels and provide a much-needed shortcut for relaying information and obtaining assistance. For example, close relationships with the involved agencies and staff were critical for the response success during the 2000 Tornado in Texas (McEntire, 2002). In contrast, ineffective relationship management can cause inter-jurisdictional tensions, exacerbate conflicts and jeopardize concerted efforts. Yeo and Comfort (2017) illustrate how polarized interagency relationships disrupted coordination and led to unprecedented damage in the 2011 Thailand floods.

H3b. Effective relationship management with relevant jurisdictional actors is positively associated with MMEO under extreme events.

Transit agencies and extreme events

Transit agencies are particularly vulnerable to disruptions and damage induced by extreme events. Extreme events not only diminish the quality and reliability of transit services, but also threaten network efficiency and mobility across interconnected and interdependent transit infrastructure. The performance of transit agencies during extreme situations furthermore has broader ramifications for local and regional response and social equity.

Several characteristics of these agencies also lend well to the study. Many transit agencies today rely on some form of outsourcing to operate transit services. They serve urbanized areas that usually cut across jurisdictional boundaries. Should an extreme event occur, the agencies depend on myriad organizations for power supply, traffic control, monitoring and maintenance of facilities, clearing of the rights-of-way and logistic support. The sector has also been increasingly aware of the importance and benefits of relationships with key actors prior to extreme events (AASHTO, 2017).

Data

We use survey data collected from US transit agencies in 2016, matched with agency profile and operational data from the National Transit database (NTD), disaster declaration data from the Federal Emergency Management Agency (FEMA), along with 2010–2015 demographic data taken from the US Census Bureau.

The study's target population includes all major US metropolitan fixed-route transit agencies with an annual fare revenue of at least one million dollars in 2013 ($N = 312$). We identified the top manager in five departments for inclusion: maintenance, operations, engineering, service planning and strategic planning. As some agencies were unreachable after repeated attempts and not every agency has all five departments or functions, the final list included 892 managers from 273 agencies.

The instrument was administered online from April 28th to June 11th, 2016 to an adjusted sample of 862 managers. A total of 306 individuals completed the survey, yielding a survey response rate of approximately 35.5%. Of the 273 agencies surveyed, 199 provided

at least one response (72.9%). The nonresponse comparison analysis showed no difference by size or region between agencies in the sample frame ($n = 273$) and the target population ($n = 312$) ($p > 0.10$).

Measurement

Dependent variable

Inspired by their measure for means efficacy of computer systems (Eden *et al.*, 2010), MMEO is measured with an index of five items. The questionnaire asked respondents: “How confident are you that your agency is able to effectively respond to the following possible occurrences? (1) Terrorism incidents; (2) disease transmission or public health concerns; (3) Social or political protests; (4) extreme weather events (e.g. major storms); (5) Natural disasters (e.g. earthquakes) (Scale 1–5: 1 = low confidence, 5 = very high confidence). The Cronbach’s alpha is 0.83 and an ANOVA test shows there is no significant difference across departments of managers’ reported level of MMEO (DF = 6, F statistic = 1.00, $p = 0.423$).

Independent variables

Organizational slack takes an organization’s annual fund received from all sources in excess of its annual expenditure in 2015. We re-centered this variable to avoid negative values and adopted its natural log. *Outsourcing* uses the ratio of contracted vehicle revenue hours (VRHs) to total VRHs an agency provided in 2013. Given most organizations have no or complete contracting, we treated it as an ordinal variable valued at 0 (no contracting), 1 (partial contracting) and 2 (complete contracting). *Political Fragmentation* is measured as the total number of cities and towns covered by each agency’s service area.

We measure an agency’s effectiveness in managing relationships with jurisdictional stakeholders with two variables: *Horizontal Relationship Management* with local stakeholders and *Vertical Relationship Management* with state or federal stakeholders. The two measures were based on a survey question asking respondents: “How effective is your agency in managing its relationship with each of the following groups?” (1) mayor; (2) mayor’s council or other elected officials; (3) city departments and agencies; (4) state governor; (5) state legislature; (6) state agencies; (7) federal government (Scale 1–5: 1 = very ineffective, 5 = very effective). Horizontal relationship management used responses to the first 3 items and vertical relationship management used the latter four. The Cronbach’s alpha for the two scales is 0.83 and 0.87, respectively.

For variables constructed from multi-item scales (i.e. MMEO, horizontal and vertical relationship management), the analysis extracted the factor score standardized at a mean of zero using Thurstone’s regression method.

Control variables

We added three categories of control variables: organizational environment, organizational attributes and individual characteristics. A primary factor in an organization’s environment is its past experience with extreme events. Increased experience can stimulate risk perception, precipitate adaptation and enhance MMEO (Miao *et al.*, 2018b; Zhang *et al.*, 2018). Alternatively, organizations exposed to repeated extreme events may find themselves increasingly unable to manage future shocks, thereby deflating MMEO. *Number of disasters* is measured by the total number of FEMA’s declarations of major disasters and emergencies for the counties covered by an agency’s primary urbanized area from 2011 to 2015. Since over 90% of those events are natural disasters, we included four Census region dummies to account for weather hazard profiles across regions. *Density* uses the natural log of population in an agency’s service area normalized by the area’s size. *Commute time* is the mean travel time to work in minutes.

Regarding organizational attributes, *bus only* is coded 1 if the agency operates buses exclusive of heavy or light rail services. *Standalone agency* is coded 1 if the agency is a special-purpose government and 0 if affiliated to a city or state government.

Finally, we considered individual characteristics. Previous research suggests that a top manager's perception and behavior is a function of their age, tenure, education level and functional specialty (Hambrick and Mason, 1984). *Age* is the respondent's age in 2016. *Tenure* is the number of years the individual has worked in the agency. *Education* measures the respondents' highest education level. We added a set of dummy variables categorizing the respondents' job function.

Table 1 reports the descriptive statistics and bivariate correlations. We see a high correlation between the number of disasters and political fragmentation. All else equal, organizations serving more jurisdictions usually cover a larger geographical area and are more likely to encounter an extreme event. Given the importance of an agency's past disaster experience, we retained this variable in analysis and did a multicollinearity test to evaluate this treatment (O'Brien, 2007).

Common method bias

We carried out a series of statistical and procedural remedies to address potential common method bias (CMB). We measured three out of five predictors using separated institutional sources. Those predictors such as organizational slack, contracting ratio and fragmentation tend to capture stable organizational status and precede the measurement of the dependent variable (Spector, 2019). The only two predictors adopted from the survey are placed physically apart from the dependent variable to minimize stylistic response or contextual effects (Podsakoff et al., 2012). The survey was pretested with 20 respondents to avoid ambiguity or confusion in the item wording (Podsakoff et al., 2003). Using Harman's test, the model shows unacceptable fit ($RMSEA = 0.199$, $CFI = 0.554$, $TLI = 0.455$, $SRMR = 0.152$) and the single factor accounts for 30% of the total variance. Given that recent simulation studies (Fuller et al., 2016) demonstrate that common method variance needs to be high (approaching 70%) to bias true relationships between variables, the concern for bias is mitigated.

Analysis and findings

Since MMEO is approximately normally distributed, we performed a multiple linear regression. We estimated a baseline model with only the control variables in Model 1 and then inserted all the key variables in Model 2 to assess their explanatory power. Both models applied robust and clustered standard errors to account for the nested data structure.

Table 2 presents the regression results, including the coefficients followed by standard errors. We conducted a joint Wald test to see if all the coefficients of the key variables are simultaneously equal to zero. The null hypothesis is rejected with $\chi^2(1, N = 5) = 91.33$, $p < 0.00$. While the 18 control variables explain approximately eight percent of the variance in MMEO, adding the five independent variables increases the adjusted R -squared to 31%. Among individual characteristics, only tenure, a variable that captures organizational embeddedness more than an individual characteristic, is significantly associated with MMEO.

Post-estimation diagnostics on Model 2 show that the residuals are approximately linear, normally distributed and have equal variance. The variance inflation factor (VIF) is largest for political fragmentation ($VIF = 6.24$) and number of disasters ($VIF = 4.05$). Both values are within the most common rule of thumb of 10 (O'Brien, 2007).

Table 1.
Summary statistics
and bivariate
correlations

	Mean	SD	Min	Max	N	1	2	3	4	5	6	7	8	9
(1) MMEO	0	0.72	-2.06	1.85	306	1.00								
(2) Slack (log)	15.84	2.34	0	22.08	304	0.26*	1.00							
(3) Outsourcing ratio	0.86	0.68	0	2	304	0.02	0.03	1.00						
(4) Political fragmentation	74.25	130.81	1	753	304	0.11	0.32*	0.03	1.00					
(5) Horizontal relationship management	0	0.53	-2.33	0.84	306	0.25*	-0.08	0.00	-0.08	1.00				
(6) Vertical relationship management	0	0.59	-2.47	1.21	306	0.47*	0.16*	0.07	0.07	0.54*	1.00			
(7) Number of disasters	5.73	19.26	0	125	304	0.09	0.14*	0.02	0.84*	-0.05	0.01	1.00		
(8) Northeast	0.13	0.33	0	1	306	0.04	0.08	0.02	0.44*	-0.03	0.12*	0.46*	1.00	
(9) Midwest	0.22	0.42	0	1	306	-0.11*	-0.01	-0.16*	-0.05	-0.03	-0.02	-0.10	-0.2*	1.00
(10) South	0.33	0.47	0	1	306	0.02	-0.18*	-0.13*	-0.12*	0.02	-0.02	-0.07	-0.27*	-0.37*
(11) West	0.32	0.47	0	1	306	0.05	0.13*	0.26*	-0.15*	0.03	-0.05	-0.18*	-0.26*	-0.37*
(12) Density (log)	7.51	1.29	-0.68	11.89	306	0.00	0.18*	0.08	0.23*	0.05	0.02	0.16*	-0.04	0.08
(13) Commute time (min)	24.42	4.71	15.3	41.7	306	0.18*	0.28*	0.19*	0.51*	-0.01	0.11	0.27*	0.11	-0.24*
(14) Standalone agency	0.74	0.44	0	1	306	-0.23*	-0.61*	0.02	-0.32*	0.06	-0.16*	-0.15*	-0.09	0.06
(15) Bus only	0.64	0.48	0	1	306	0.02	0.33*	-0.01	-0.03	-0.06	0.04	-0.08	0.17*	0.06
(16) Executive leadership	0.17	0.38	0	1	306	-0.02	-0.08	0.02	-0.01	-0.02	-0.02	-0.02	0.01	-0.05
(17) Operations	0.29	0.45	0	1	306	-0.05	-0.08	-0.02	-0.03	0.12*	0.00	0.01	0.00	0.03
(18) Maintenance	0.17	0.38	0	1	306	0.06	-0.01	-0.05	-0.04	-0.04	-0.03	-0.02	-0.02	0.00
(19) Planning	0.23	0.42	0	1	306	0.03	0.11	0.07	0.03	-0.11*	-0.05	0.01	-0.02	-0.06
(20) Engineering	0.04	0.19	0	1	306	0.06	0.10	0.02	0.10	-0.10	-0.02	-0.04	0.02	0.01
(21) Safety	0.05	0.21	0	1	306	0.04	0.02	-0.05	-0.04	0.09	0.21*	-0.02	0.01	0.07
(22) Other positions	0.06	0.23	0	1	306	-0.10	0.01	-0.01	0.05	0.08	0.01	0.07	0.03	0.07
(23) Education	4.03	0.94	1	6	304	-0.06	0.17*	0.10	0.08	0.00	-0.03	0.06	-0.10	-0.08
(24) Age	51.73	10.1	23	76	292	0.06	-0.01	0.00	0.01	-0.01	-0.02	-0.04	-0.04	-0.06
(25) Tenure	13.36	10.39	0	42	303	0.1	0.04	-0.1	0.13*	-0.05	-0.06	0.1	0.06	0.05

(continued)

	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
(10) South	1.00															
(11) West	-0.48*	1.00														
(12) Density (log)	-0.18*	0.13*	1.00													
(13) Commute time (min)	-0.09	0.23*	0.16*													
(14) Standalone agency	0.11	-0.10	-0.22*	1.00												
(15) Bus only	-0.26*	0.09	-0.08	0.10	1.00											
(16) Executive leadership	0.15*	-0.11	-0.08	0.05	-0.06	1.00										
(17) Operations	-0.01	-0.01	0.01	-0.09	-0.05	-0.29*	1.00									
(18) Maintenance	-0.02	0.03	0.01	-0.07	-0.03	-0.21*	-0.29*	1.00								
(19) Planning	0.04	0.03	0.05	0.06	-0.06	-0.24*	-0.34*	-0.25*	1.00							
(20) Engineering	-0.10	0.08	0.02	0.06	-0.15*	-0.09	-0.13*	-0.09	-0.11*	1.00						
(21) Safety	-0.02	-0.05	-0.01	-0.03	0.02	0.07	-0.10	-0.14*	-0.10	-0.12*	-0.04	1.00				
(22) Other positions	-0.14*	0.06	0.01	0.07	-0.01	-0.04	-0.11*	-0.16*	-0.11*	-0.13*	-0.05	-0.05	1.00			
(23) Education	0.10	0.05	0.18*	0.08	-0.17*	0.00	0.16*	-0.07	-0.30	0.25*	-0.04	-0.03	-0.03	1.00		
(24) Age	0.00	0.08	0.00	0.10	-0.03	-0.02	0.00	0.12	0.06	-0.14*	-0.02	0.01	-0.06	-0.09	1.00	
(25) Tenure	-0.11	0.02	0.07	0.09	-0.06	-0.01	-0.02	0.11	-0.04	-0.06	0.06	-0.07	0	-0.22*	0.4*	1
Note(s): * $p < 0.05$																

Table 1.

	Model 1	Model 2
Organizational slack		0.063 (0.023)**
Contracting ratio		0.013 (0.058)
Political fragmentation		−0.001 (0.001)#
Horizontal relationship management		0.096 (0.08)
Vertical relationship management		0.419 (0.067)***
Number of disasters	0.003 (0.003)	0.01 (0.004)*
Midwest	−0.069 (0.174)	0.028 (0.149)
South	0.062 (0.176)	0.141 (0.153)
West	0.031 (0.168)	0.126 (0.147)
Density (log)	−0.032 (0.046)	−0.043 (0.044)
Commute time (min)	0.019 (0.009)*	0.015 (0.01)
Bus only	−0.367 (0.098)***	−0.117 (0.12)
Standalone agency	−0.081 (0.101)	−0.117 (0.097)
Operations	0.042 (0.12)	−0.078 (0.101)
Maintenance	0.148 (0.136)	0.094 (0.114)
Planning	0.153 (0.126)	0.119 (0.109)
Engineering	0.097 (0.212)	0.177 (0.224)
Safety	0.247 (0.201)	−0.108 (0.16)
Other positions	−0.27 (0.184)	−0.374 (0.032)*
Education	−0.078 (0.048)	−0.068 (0.044)
Age	0 (0.005)	0 (0.004)
Tenure	0.062 (0.176)	0.009 (0.004)*
R-squared	0.124	0.361
Adjusted R-squared	0.069	0.308
# of parameters	17	22

Table 2.
Determinants of top
manager confidence
in organization

Note(s): 290 observations in 189 clusters; # $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Reference category for position dummies: Executive Leadership
Reference category for region dummies: Northeast

As a robustness check, we carried out additional analyses to account for potential variability in agencies' slack levels. The agencies' slack levels were highly consistent in the five years prior to our data collection (2011–2015), with a year-to-year average correlation of over 0.90. We re-estimated the full model replacing the 2015 slack measure with a 3-year (2013–2015) and 5-year (2011–2015) average, respectively. The 3-year average slack predicts MMEO at a significance level of 0.05 ($\beta = 0.039$, $p = 0.046$) and the 5-year average slack predicts MMEO at a significance level of 0.10 ($\beta = 0.035$, $p = 0.095$), both of which confirm the positive effect of organizational slack on MMEO.

Hypothesis 1 postulates a positive relationship between organizational slack and MMEO. The analysis supports this hypothesis. More auxiliary resources enhance operational and strategic flexibility and widen response repertoire. It also empowers managers to make anticipatory adjustments to build capacity, fostering a stronger sense of MMEO.

Hypothesis 2 expects that an organization's level of outsourcing is negatively associated with MMEO. The coefficient is not significant. This might be due to the bimodal distribution of the outsourcing variable, which limits its ability to detect continuous variation in the dependent variable, especially when the sample size is small and underpowered (Royston *et al.*, 2006).

Hypotheses 3a and **3b** focus on political embeddedness. **Hypothesis 3a** expects a negative relationship between political fragmentation and MMEO. Results marginally support this hypothesis. Even though the coefficient is small, cumulative effect of political fragmentation is non-trivial because the transit organizations serve an average of 74 cities and towns.

Hypothesis 3b expects effectiveness in managing both horizontal and vertical relationship to boost MMEO. Only vertical relationship management turns out to be a significant and positive predictor for MMEO, partially supporting **Hypothesis 3b**. The non-significant effect of horizontal relationship management may be attributed to the broad span of transit service areas. Compared with vertically linked networks, horizontally linked networks also suffer from more ambiguous and indirect interdependence among involved organizations. The finding highlights the importance of higher-level governments to support crisis response.

Discussion

Previous research on top managerial efficacy has narrowly featured managers' self-efficacy. This study draws on Eden's theory of means efficacy and extends the concept of managerial efficacy to include efficacy beliefs deriving from external means. We focus on MMEO as a primary form of MME. Recognizing that tasks vary in the degree of means-dependence, we anchor the investigation of MMEO in high means-dependent contexts: extreme events.

The framework integrates theories of planned behavior and resource dependence to explain how MMEO is shaped by factors both internal (i.e. perceived organizational capacity) and external (i.e. perceived controllability) to the organization. Internally, it is linked to a top manager's perceived organizational capacity to meet the demands during extreme events; externally, the level of outsourcing, fragmentation of the political setting and interactions with key jurisdictional stakeholders influence managers' perceived controllability. The hypotheses are partially supported: organizational slack and vertical relationship management are positively associated with MMEO, while political fragmentation is marginally and negatively associated with MMEO.

The study demonstrates that MMEO is jointly determined by an organization's internal competence, as well as its embeddedness in and interaction with the external environment. The identified determinants of MMEO can be targeted for intervention strategies to develop managerial efficacy. The findings point to the importance of developing organizational slack and engaging in relationship building with key actors. We also find political fragmentation is negatively associated with MMEO. As organization leaders cannot redefine or alter their service area, it would be more productive to focus on domains over which they have control, such as growing organizational slack and engaging in pre-event relationship building with stakeholders.

We acknowledge a few limitations. Lacking performance outcome data, we relied on extensive existing studies to establish the importance of MMEO. Future research will need to examine the influence of MMEO on organizational response during extreme events. Second, the focus on transit agencies may limit the applicability of the findings. However, the use of a national representative survey, coupled with the similarity of transit agencies with many resource intensive organizations, including utility, power and sewage management, helps attenuate this concern. Additionally, the transit sector is also a mixed service industry with both public and private sector features and moderate slack needs, further bolstering the study's generalizability. Third, the study could benefit from including measures on managerial self-efficacy to investigate its linkage and interaction with MMEO (Hannah *et al.*, 2008). We are also cognizant of the possible negative effects associated with managerial efficacy beliefs, such as excessive optimism or risk underestimation. However, in general, negative emotions and cognition are more prevalent and detrimental to organizational performance under high-stress situations (Luthans and Church, 2002). Finally, since, managers form and update efficacy beliefs through patterned behavior of organizational members and common problem-solving experience, our framework may have limited applicability to novel extreme events with no *a priori* evidence of performance or unknown courses of action.

Conclusion

This study makes a few contributions. First of all, drawing on Eden's theory of means efficacy (Eden, 2001), it advances the management and leadership research through orienting attention to a critical but oftentimes neglected aspect of managerial efficacy: external sources of managerial efficacy (Hannah *et al.*, 2008). It thus enables a broader concept of managerial efficacy, incorporating not only managers' perceived internal resources to act and get things done, but also their perceived abilities to orchestrate *external* means to produce desired outcomes. This broader perspective matches up with the complex and dynamic contexts in which managers typically find themselves. The importance of considering the external sources of managerial efficacy is reinforced by the complexity and intractability of today's problems highlighting means-dependence for achieving organizational goals.

We also believe this study brings conceptual clarity to the notion of "confident manager" or "managerial confidence" widely applied in management and leadership scholarship. Extant studies include a mix of referents for managerial confidence, ranging from their internal resources such as learning, intelligence and expertise (Appelbaum and Hare, 1996; Bandura, 2000; Brockner and James, 2008), their team, subordinates or clients (Fusco *et al.*, 2015; Ladegard and Gjerde, 2014) to their organization (Paglis and Green, 2002; Zhang *et al.*, 2020). Given that internal and external sources of efficacy beliefs have different referents and likely disparate determinants and implications, the introduction of distinct external sources of managerial efficacy opens up avenues for theoretical and practical applications not implied in previous managerial efficacy literature.

The study goes beyond the theory of means efficacy to explore antecedents of MME, using MME of the organization as a vehicle for examination. Integrating the theory of planned behavior, it develops a theoretical framework conceptualizing MMEO along two dimensions: perceived organizational capacity and perceived controllability. Although we select extreme events as an anchor for a high means-dependent task environment due to research design considerations, those relationships should hold in other general settings with lower means-dependence. Drawing on Ajzen's (1991) and Eden's theory (2001), the concept of MMEO and its associated relationships would be valid and important in situations that involve getting things done in an interdependent system where handling contingencies created by events and actors beyond an organization's boundary is a necessity (Hickson *et al.*, 1971). Examples include procurement, marketing, contracting, supply chain management, R&D, data sharing and integration. On the other hand, we also note boundary conditions for MMEO. Eden *et al.* (2010) state that the importance of means efficacy hinges on the indispensability of tools for achieving successful performance. In tasks where external means play only a minimal role, such as those involving managerial cognitive calculation or learning capability, MMEO would be relatively unimportant. In reality, however, today's organizations in all sectors and fields are increasingly involved in transaction with myriad external actors and are constrained by a web of interdependent relationships that influence performance outcomes (Pfeffer, 1992), all of which serve to increase the applicability and importance of MMEO.

Moreover, the study takes a contextual focus on extreme events, directly responding to the call for taking greater account of contexts in leadership theories (Porter and McLaughlin, 2006). It effectively combines insights from the macro-level organizational theory and brings in the essential contexts to inform understanding and prediction of micro-level cognition. The contextualization in extreme events further bridges the organizational behavior and crisis management literature. In contrast with most extant studies with a predominant focus on managerial vulnerabilities and weaknesses, the study underscores MMEO as a critical attenuator amid the extreme situations (Luthans and Church, 2002). The focus on psychological strengths and capacities provides a new focal point for studying leadership and organizational response in times of crises.

The study raises several new questions for further work. This study confines us to investigate antecedents of only one form of MME – one informed by the organization. To what extent does the explanatory framework of MMEO apply to MME derived from other forms of means, such as teams and work groups? The framework suggests that both perceived organizational capacity and perceived controllability can affect MMEO during extreme events. What other contextual factors in those categories predict MMEO? Future work should incorporate a wider range of indicators for an organization's task-specific capacity and embeddedness in the external environment. Finally, we note the potential interaction between self-efficacy and means efficacy. In this study we choose extreme events as the anchor to study MMEO, because in such high means-dependent contexts, MMEO can overshadow self-efficacy in determining task performance. However, is the opposite true that in low means-dependent tasks, self-efficacy can overshadow MMEO in an organizational environment? Or could the two be difficult to disentangle? Future work is invited to examine the interplay of those two forms of managerial efficacy under varied levels of means-dependence.

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